

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

ME-5005(2)-CBGS

B.E. V Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Metrology and Inspection

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Draw neat diagram wherever required.

1. a) Give the structure of generalized measuring system and explain. Give the difference between accuracy and precision with an example.
b) Briefly explain the modes of operation and applications of the measurement system.
2. a) What are the considerations for deciding the limit on the limit gauge?
b) Explain the principle of interferometer by suitable diagram.
3. a) How are major and minor diameters of thread measured?
b) Explain the following screw thread effective diameter method:
 - i) One wire method
 - ii) Two wire method

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[2]

4.
 - a) Explain the principle of measuring gear tooth thickness by base tangent method.
 - b) Describe optical bevel protractor and explain how it is used to measure the angle.
5.
 - a) Describe a method of determining an absolute length of slip gauges using interferometer.
 - b) Explain in details the roundness testing machine.
6.
 - a) Explain various systematic and random errors in measurements.
 - b) What is Interference band? Explain flatness interferometers with neat sketch.
7.
 - a) What is comparator? What is their use? Explain different types of comparators.
 - b) With neat sketch explain the construction and working principle of differential pneumatic comparator.
8. Write short notes on following (any two):
 - a) Advanced measuring machines
 - b) Laser vision
 - c) 3D metrology

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